ECG Arrhythmia Classification and Model Explainability via Permutation Feature Importance

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INTRODUCTION

Cardiovascular diseases remain one of the major causes of mortality worldwide. One of the ways to diagnose them is the **Electrocardiogram (ECG)**, which is considered as a gold standard because it is a non-invasive method for monitoring the electrical activity of the heart and detecting conditions such as arrhythmias. An ECG records the heart's electrical activity as a characteristic sequence of waves: the **P-wave** (atrial depolarisation), **the QRS complex** (ventricular depolarisation) and **the T-wave** (ventricular repolarisation). Cardiologists detect arrhythmias by comparing these waves with the normal cardiac cycle. Manually analyzing a 24 hours ECG is time consuming and prone to human error. Additionally, analyzing this data and generating the annotated data is very toilsome.

This has motivated the use of **machine learning** to learn the features of a heartbeat and classify it automatically.

However, using machine learning for this purpose also has its own limitations:

1. **Unrealistic performance estimates:** A common evaluation protocol shuffles heartbeats from all patients and splits them randomly into training and test sets. Beats from the same patient share characteristics such as electrode placement, pre-existing conditions and medication, so beats of the same class from one patient look alike. Evaluating on patients the model has never seen gives a more honest estimate of how it would perform in practice.
2. **Lack of transparency:** Many classifiers behave as black boxes, so clinicians cannot check whether a prediction rests on genuine electrophysiological features or on spurious artefacts such as noise, baseline wander or motion, which makes them reluctant to trust the model.
3. **Lack of data:** Annotating each second of each heartbeat is not an easy task. It is very time-consuming and requires dedicated people to do it.

This report addresses the first 2 problems of using the **MIT-BIH Arrhythmia Database**. The database contains ambulatory ECG recordings from 47 subjects, digitized at 360 samples per second and annotated by cardiologists. Each beat is a 275-point segment centered on its **r-peak** and standardised and is labelled with one of eight classes: [N, L, R, V, A, F, f, /].

These classes are heavily imbalanced. Most beats are normal heartbeats and belong to the class N.

Clear Statement of Predictive/Exploratory Question

The study asks two questions:

1. **Predictive:** How accurately can the classifiers classify the eight heartbeat types under 2 different data splitting protocols (beat hold-out and patient hold-out).
2. **Exploratory:** Which part of the heartbeat (i.e. the P wave, the QRS wave or T wave) does the classification rely on? Which part carries the most weight in the classification, as measured by **permutation feature importance**?

We answer these questions by training both classifiers under each validation protocol and evaluating them with accuracy, precision, recall, F1 score and confusion matrices.

We also apply 5-fold permutation feature importance to each model.

Methods

Dataset and Signal Pre-processing

Data source:

- The dataset used here is the **MIT-BIH Arrhythmia Database.**
- It includes ambulatory ECG recordings from 47 subjects, digitised at 360 samples per second with 11 -bit resolution over a 10-mV range on two channels.
- These ambulatory recordings capture intermittent, spontaneous arrhythmias that occur unpredictably.
- However, they also introduce real-world noise: motion artefacts, baseline drift (from respiration), and muscle tremors (electromyographic noise).

48 half-hour recordings were taken from **47 subjects** (Records **201** and **202** are from the same patient). Approximately 60% were inpatients; 23 recordings were chosen at random, and 25 were selected intentionally because they contained clinically significant, uncommon arrhythmias (such as ventricular tachyarrhythmias, heart blocks, and pacemakers).

The recordings were annotated beat by beat by cardiologists. The signals were then read with the Python package **'wfdb'** (Waveform Database), which handles the record files (.hea, .dat, .atr) and beats from both channels were extracted and stacked so that all beats in each record were obtained.

Two channels include:

1. **Upper channel:** modified limb lead II (in most records), obtained by placing the electrodes across the chest along the electrical axis of the heart. It produces tall, upright R-peaks and distinctive P-waves
2. **Lower channel:** Electrodes are placed over specific intercostal spaces to view horizontal ventricular conduction.

Sampling rate: The analogue electrical voltage from the body is sampled 360 times every second, which equals 2.78 milliseconds per sample.

Beat Segmentation

Each ECG cycle has the waves P, Q, R, S and T. The R peak is a tall, sharp upward spike in the middle of the cycle, and it represents the peak electrical surge of ventricular depolarisation.

It has the steepest slope and the largest amplitude in the entire heartbeat.

The average human heart beats around 60-80 times per minute at rest, so the average beat duration is about 0.8 seconds. A 275-sample window corresponds to about 764 ms. This is long enough to capture all phases of an individual cardiac cycle and at typical resting rates short enough not to overlap with the next P-wave.

Standard machine learning classifiers such as SVM cannot process variable-length inputs; they need fixed-dimensional matrices.

If beats were not aligned to the R-peak, column 50 might represent a P-wave in one patient, a QRS-peak in another, and a flat baseline in a third. The machine learning model would then be unable to learn consistent patterns.

By centring every beat on the R-peak, each column index j acquires a consistent physical meaning:

Time offset from R-peak = (j − 137) × 2.78 ms (to within one sample)

After this, the distance calculations compare exactly same physiological phase across thousands of beats.

The waves therefore fall into different time frames, as shown below:

Column indices	Time relative to R-peak	ECG feature
0 - ~100	-380 ms to -100 ms	P-wave and PR segment
~100 - ~175	-100 ms to +100ms	QRS Complex
~175 - 274	+100 ms to +380ms	ST segment and T-wave

Standardisation

- All the beats are standardised to zero mean and unit variance, so that the DC offset and amplitude differences between recordings are removed and the slow baseline shifts caused by the patient's breathing and chest expansion are reduced.
- There are various factors, such as skin impedance and sweating (dry skin conducts electricity differently from moist skin) and anatomy and body composition (a thin patient or an athlete with an enlarged heart will produce larger R-peaks than an elderly person) that change the signal amplitude between patients.
- The class ID and patient record numbers were appended at the end of each beat so that patient origins never become misaligned from their corresponding signals during shuffling or sorting.
- So each row has 275 signal values, 1 class ID and 1 patient ID.

Class Selection and Encoding:

Only the original eight annotation classes were kept , all other annotations were removed, and the classes were encoded as numbers from 1 to 8.

Annotations	Beat Type	Class ID
N	Normal	1
L	Left bundle branch block	2
R	Right bundle branch block	3
V	Premature ventricular contraction	4
A	Atrial premature beat	5

F	Fusion of ventricular and normal beat	6
f	Fusion of paced and normal beat	7
/	Paced beat	8

As the data were imbalanced, the class count was as follows:

Normal beats (N) have ~114368 samples (90% of data)
Fusion beats (F) have ~ 873 samples (<0.7 % of all data)
Imbalance ratio - over 136:1

Such a model would achieve a high accuracy of 90% or more, but would have a poor sensitivity to PVCs and fusion beats.

Instead of standard naive balancing, we use **down sampling** of the majority classes and **up sampling** of the minority classes.

This ensures every class is equally present in the data, forcing the decision boundaries to give equal weight to every pathology. The training data were resampled, but the test data was never resampled.

The reasons why the test data were never resampled:

1. If we resampled the entire dataset before splitting, identical bootstrap duplicates of rare minority beats would end up in both the training set as well as the testing set. The model would then simply memorise the duplicate test beats present in the training data and overfit the model.
2. Keeping the test set untouched evaluates how the model will perform in the **real world**, where it must successfully detect rare pathological beats without raising false alarms on the far more numerous normal beats.

Validation Protocol

In machine learning, models typically assume that the training data are independent and identically distributed.
But in healthcare settings such as ECG, the heartbeats of the same person are highly dependent. A person's heartbeat shares the same electrical properties, anatomical structure and electrode placement. So, the way we partition this data determines the output of the model.

Here we have chosen 2 methods:

Beat Hold-out (Intra patient Split)

- All the heartbeats from the 48 records are pooled in a single CSV file and randomly shuffled.
- These randomly shuffled beats are then split 75% for training and 25% for testing.
- For example, about 75% of the beats of patient 101 end up in the training set and 25 % in the test set. The classifier then does not need to learn the universal medical definition of the disease; it can simply memorise the signature of patient 101.
- During testing it recognises the exact waveform shape seen in training, which inflates the accuracy. If deployed in a hospital on a new patient, such a model may fail catastrophically because it never learned generalisable pathology; it learned patient memorisation .

Patient Hold- out (Inter-Patient Split)

- The dataset is split strictly based on the **patient ID**.
- 43 records from the training set (resampled to balance the classes).
- Five specific patients (104, 113, 119, 208 and 210) are held out for testing.
- Patient hold-out evaluates true out-of-sample generalisation across differing human anatomies, lead placements, and ages.
- However, the held-out patients contain no L,R or A beats, so these classes cannot be assessed on the test set.

Classification Models and Hyperparameters

Support Vector Machine (SVM)

- The support vector machine finds the decision boundary that separates the classes with widest possible margin.
- For non-linear problems such as ECG morphology, a kernel function implicitly maps the beats into a higher-dimensional space in which they can be separated.
- We use the Radial Bases Function (RBF) kernel, which depends on the Euclidian distance between two beats; because this distance is sensitive to scale, the per-beat standardisation described above is important.
- Multi-class classification is handled by scikit-learn with a one-versus-one scheme.
- The two important hyperparameters are **C** and **gamma**.
- C is the penalty for misclassified training beats: a small C gives a wider margin and tolerates more training errors, whereas a large C fits the training data more tightly.
- Gamma sets how far the influence of a single training beat reaches: a large gamma gives a wiggly, local decision boundary that can overfit, whereas a small gamma gives a smoother one.

Random Forest (RF)

- A random forest is an ensemble of decision trees.
- Each tree is trained on a bootstrap sample of the training beats and at every split, considers only a random subset of the 275 time points.
- The forest predicts the class chosen by majority of its decision trees.
- Averaging many decorrelated trees reduces variance and makes the model less prone to overfitting than a single tree and no feature scaling is required.
- The important hyperparameters are the number of trees (n_estimators), the maximum depth of each tree (max_depth), the minimum number of samples needed to split a node or form a leaf and the number of features considered at each split (max_features).

Hyperparameter Settings and Training

Both models were trained on the class-balanced training set of each validation protocol, giving four trained models and each model was evaluated once on the corresponding untouched test set. The hyperparameter values used are listed below:

Model	Hyperparameter	Value used
SVM	Kernel	RBF
SVM	C	10
SVM	gamma	Scale
Random Forest	n_estimators	150
Random Forest	max_depth	15
Random Forest	max_features	sqrt (default)

Evaluation Metrics

For each class,

Metric	Description
Precision	fraction of beats predicted as that class that truly belong to it
Recall	fraction of beats of that class that were correctly identified
F1 score	Harmonic mean of precision and recall
Macro average	Takes the unweighted mean over classes, so rare classes count as much as the normal class
Confusion Matrices	are normalised by true class, so each row shows the proportion of that class assigned to each predicted class

- Models were evaluated only on the test set, using a common function that reports accuracy, precision, recall, F1 score and the confusion matrix.
- Precision, recall and F1 are reported per class and as macro averages, because accuracy alone is misleading when the test set is dominated by normal beats.
- Classes missing from the patient hold-out test set (L, R, A) were excluded from the averages and handled with explicit class labels and 'zero_division=0'.

Permutation Feature Importance:

Permutation feature importance is used to find out which parts of a heartbeat a trained model depends on. The idea is to scramble one part of the input and see how much the predictions get worse. If performance drops a lot, the model was relying on that part, and if it barely changes, the part matters little.

Each beat has 275 time points, which is too many to interpret one by one. We group them into 11 non-overlapping slices of 25 samples (69.4 ms) each and treat each slice as a single feature.

- Slice 1 covers samples 0-14
- Slice 11 covers samples 250 - 274

Because the R-peak sits in the centre of each beat, slice 6 (samples 125 - 149) contains it.

Working with slices helps us link the results to the P-wave, the QRS complex and the T-Wave and also keeps the computation tractable.

Procedure

1. Split the data into K = 5 folds, stratified on the beat class.

2. For each fold k, fit a new model on the other four folds and predict on the held-out fold to obtain the baseline predictions.

3. For each slice j = 1, ..., 11, copy the held-out beats and randomly shuffle the 25 time points inside slice j, independently for each beat. Predict again with the same fitted model. The model is not refitted for each slice.

4. For each beat i, the importance of slice $\boldsymbol{j}$ is the change in error:

$$M_{ij} = 1\left[\hat{y}_i^{(j)} \neq y_i\right] - 1[\hat{y}_i \neq y_i]$$

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where $\hat{\boldsymbol{y}}_{\boldsymbol{i}}$ is the original prediction and $\hat{\boldsymbol{y}}_{\boldsymbol{i}}^{(\boldsymbol{j})}$ is the prediction with slice $\boldsymbol{j}$ shuffled.

5. Average M_{ij} over all beats and all folds to obtain one score per slice. Scores can be read as the increase in misclassification rate caused by shuffling that slice.

We measured error as a simple 0/1 misclassification rather than the squared error in the original formulation. The class labels are categories, so squared error on the integer IDs would be meaningless, because class 8 is not "further" from class 1 than class 2 is.

With one-hot labels, the squared error of a wrong prediction is always 2 and of a correct one is 0, so the 0/1 error carries the same information.
Shuffling inside a slice was chosen over setting it to zero or to its mean, because shuffling keeps the amplitude values but destroys the shape of the wave. The model therefore loses the information in the local morphology without being shown an artificial flat segment.

Implementation and Reproducibility

All experiments were implemented in Python in Jupyter notebooks. The recordings were read with wfdb, R-peaks were detected with biosppy and the classifiers, resampling, metrics and cross-validation come from scikit-learn, with NumPy and matplotlib for numerical work and plotting. The notebooks are run in this order:

1. data pre-processing
2. splitting and resampling
3. classification and permutation feature importance

A fixed random seed ([seed value]) was used for the data split, the resampling and the models so that the results can be reproduced.

Results / Metrics

Experimental Setup and Rationale

Four experiments were run, one for each combination of classifier (SVM, Random Forest) and validation protocol (beat hold-out, patient hold-out). Each classifier was trained on the class balanced training set of its protocol and evaluated once on the untouched test set. The tests were never resampled, so they reflect the natural class imbalance. Permutation feature importance was then computed for each of the four configurations with 5-fold cross-validation, stratified on the beat class.

Protocol	Training data	Test data
Beat hold-out	Random, 75% of all beats from the 48 records, resampled to equal class sizes	Remaining 25% of beats unsampled
Patient hold-out	All beats from the 43 training records, resampled to equal class sizes	All beats from patients: 104, 113, 119, 208 and 210; not resampled.

The rationale for the experiments is as follows:

- **Beat hold-out** serves as the baseline; it shows how well the models separate the classes when the patient is not controlled for.
- **Patient hold-out** measures the generalisation to patients never seen in training.
- **Comparing the two** quantifies how much of the beat hold-out performance comes from patient specific memorisation rather than learned pathology.
- **Permutation feature importance** shows which part of the beat each trained model relies on, comparing the SVM with the random forest checks whether the explanation depends on the model.

Classification Performance

The diagnostic classification performance of the support vector machine and the random forest was evaluated under both the beat hold-out and patient hold-out validation protocols on unseen dataset.

Model	Validation Protocol	Test Accuracy	Macro Precision	Macro Recall	Macro F1	Weighted F1	Training Time (s)
SVM (RBF)	Beat-Holdout	93.87%	0.7744	0.9530	0.8212	0.9496	0.81
Random Forest	Beat-Holdout	93.79%	0.7529	0.9493	0.8126	0.9468	1.37
SVM (RBF)	Patient-Holdout	94.34%	0.5780	0.5540	0.5601	0.9521	0.90
Random Forest	Patient-Holdout	94.02%	0.6650	0.6180	0.6342	0.9485	1.40

Table 1. Test performance of the SVM and Random Forest classifiers under the beat-holdout and patient-holdout validation protocols.

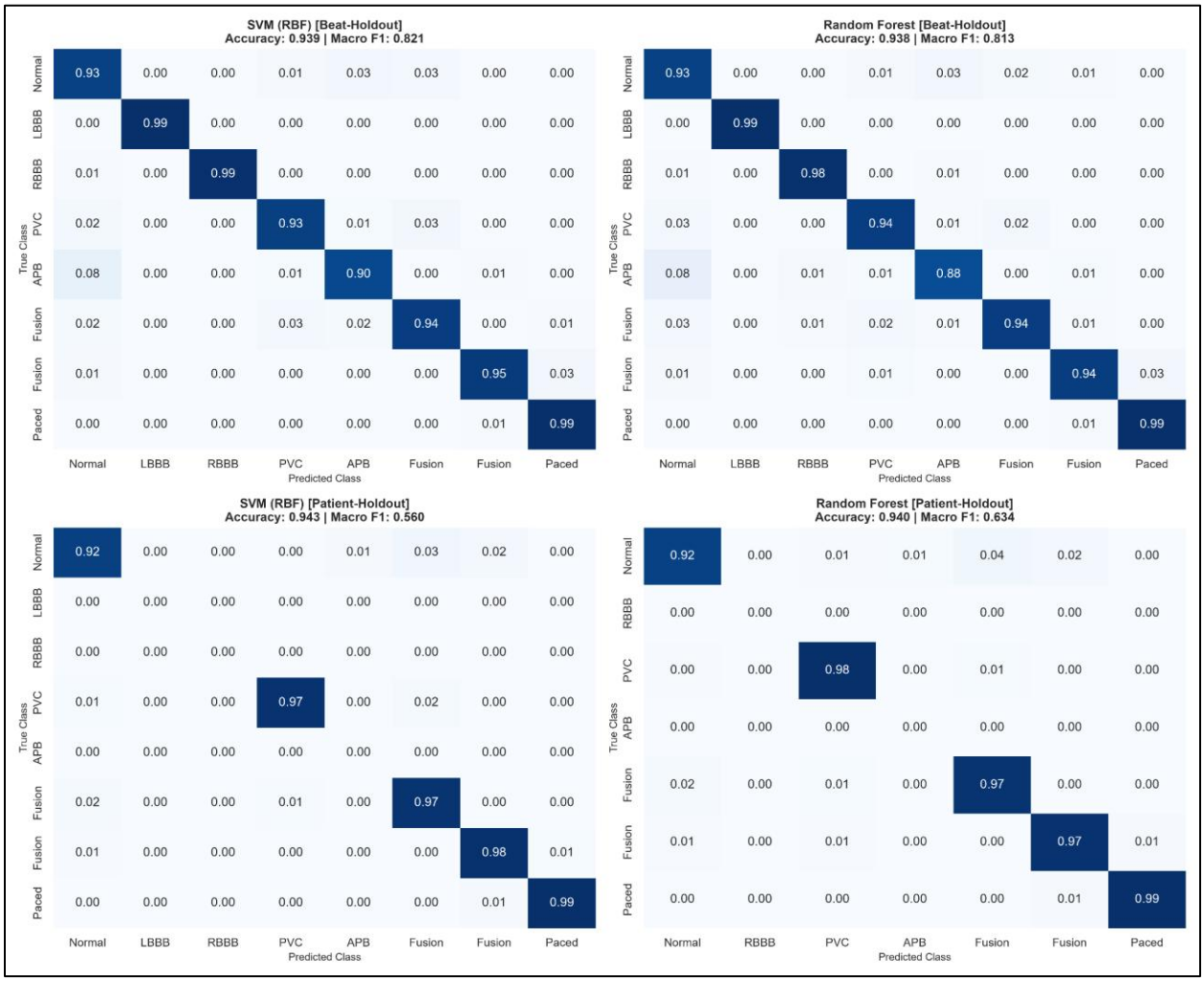
From the metrics above:

- Accuracy is slightly higher for patient hold-out (94.3 %). This appears to contradict what is stated above, but the macro F1 score tells the real story.
- Macro recall drops from 0.95 to 0.55 and macro F1 drops from 0.82 to 0.56 for the SVM.
- Accuracy is highly dominated by the normal beats (N) class.
- Majority class dominance: The five held-out patient records (104, 113, 119, 208, 210) comprise 14,482 test beats, of which over 82% belong to normal sinus rhythm (64.7%) and paced beats (17.3%). Both classifiers achieve near-perfect classification (>98%) on these distinct, high-voltage morphologies. Because overall accuracy is sample-weighted, high concordance on these dominant classes masks diagnostic failure on other rhythms.
- A critical methodological consideration in patient-level cross-validation is the definition of the macro-average denominator. In our implementation, scikit-learn metrics were parameterised with zero_division=0 and explicit class labels. Because classes L, R, and A have zero ground-

truth support (N=0N=0N=0) in the held-out patient cohort, assigning an automatic 0.0 score across an 8-class denominator imposes an artificial penalty for diseases not clinically present in the test cohort. Conversely, evaluating macro averages across the 5 actively present classes (N, V, F, f, /) isolates true cross-subject generalisation on observed pathologies.

Confusion Matrices

The 8 x 8 confusion matrices below display normalised true-positive rates (recall) along the main diagonals and reveal specific misclassification pathways across all eight arrhythmia classes.



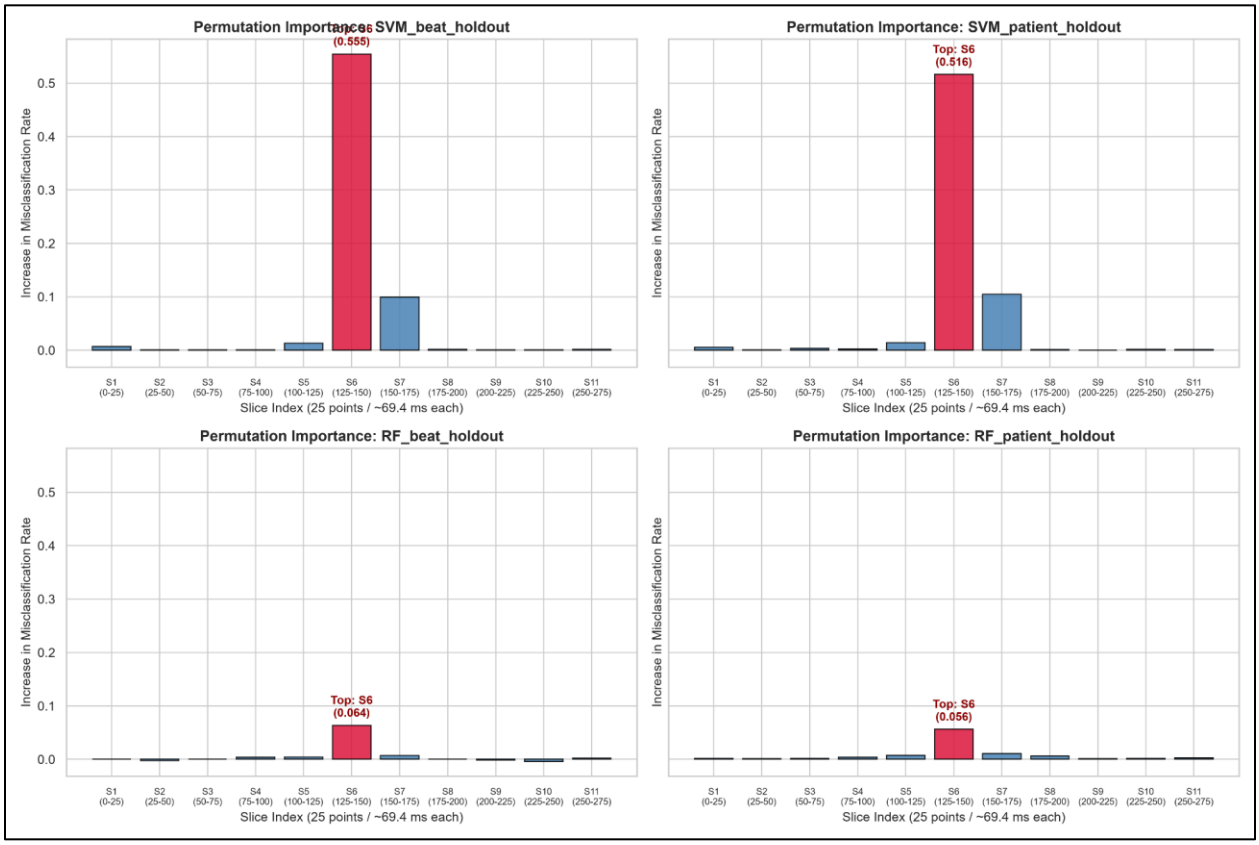
Top Row (Beat-Holdout)

Both SVM (left) and Random Forest (right) display strong diagonal dominance across all 8 classes (>90%>90\%>90% recall on most categories), demonstrating high sensitivity when beats from all patients are represented in both partitions.

Bottom Row (Patient-Holdout)

This row demonstrates the effect of unseen patient morphologies. While Normal (N), PVC (V), and Paced (/) beats retain high diagonal recall (>96%>96\%>96%), classes L, R, and A have zero test support in the held-out patients (104, 113, 119, 208, 210), and rare fusion beats suffer cross-class leakage into PVCs.

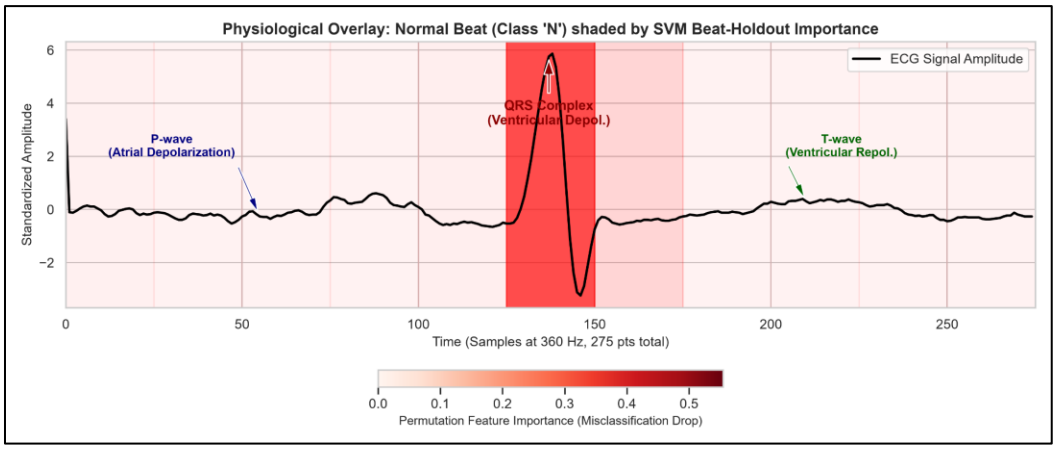
Time Slice Permutation Feature Importance (PFI)



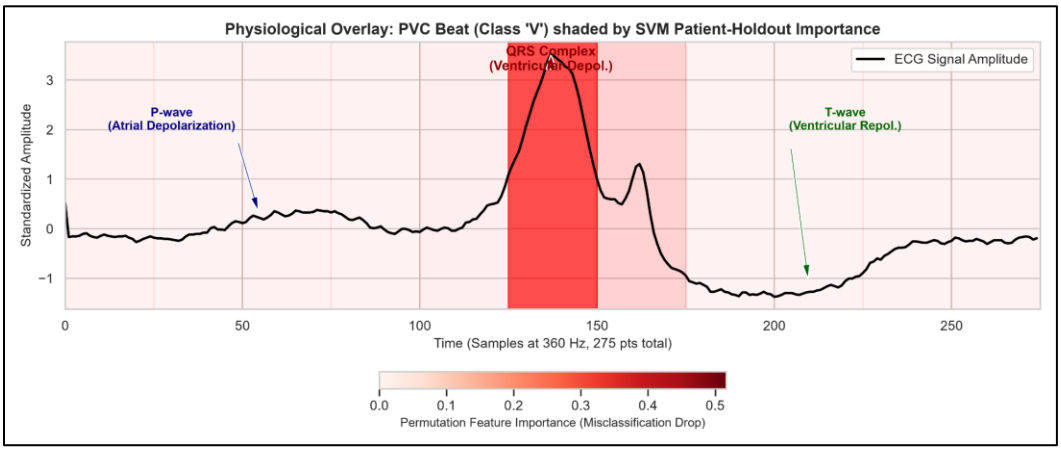
Overwhelming QRS dominance: In all four configurations, Slice 6 (samples 125–149) is the strongest predictor, followed immediately by Slice 7 (samples 150–174).
Classifier architecture contrast: The SVM demonstrates a strong, localized dependency on Slice 6 (>50%>50\%>50% error increase), whereas the Random Forest exhibits a smoother profile (~6% error increase) that distributes weight across adjacent slices.

Physiological Waveform Overlays Shaded by Importance

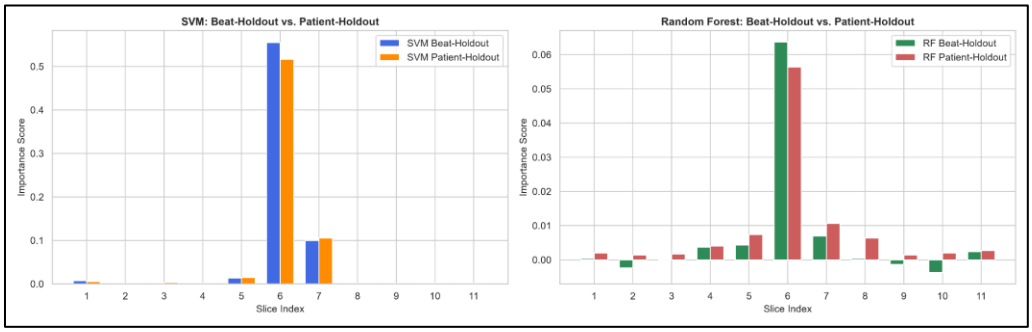
These figures project the empirical PFI scores directly onto the raw ECG signal, highlighting the correspondence between algorithmic importance and clinical cardiac landmarks.



High importance (red shading) tightly envelops the sharp, narrow R-peaks.



High importance spans the entire widened, bizarre ectopic QRS complex.



Side by side comparison of PFI between SVM and RF

Model Agreement and Spearman Rank Correlation

To evaluate whether the learned importance scores reflect genuine physiological signals rather than random artefacts of a particular classifier, we computed pairwise Spearman rank correlation coefficients across the 11-dimensional importance vectors.

Comparison Pair	Spearman Rank Correlation (rs)	Unadjusted p-value	Correlation Interpretation
SVM: Beat-Holdout vs. Patient-Holdout	0.6682	0.0246	Moderate positive rank agreement
RF: Beat-Holdout vs. Patient-Holdout	0.8856	0.0003	Strong positive rank agreement
Beat-Holdout: SVM vs. Random Forest	0.7189	0.0127	Moderate-to-strong positive rank agreement
Patient-Holdout: SVM vs. Random Forest	0.7414	0.0090	Moderate-to-strong positive rank agreement

Model Concordance and Architectural Stability Across Validation Splits

Comparing the 11-slice Permutation Feature Importance profiles reveals critical insights into how each classifier navigates patient-level variation:

1. Random Forest Stability ($r_s = 0.8856, p = 0.0003$)

The Random Forest's temporal importance distribution remains remarkably consistent regardless of whether it is trained under Beat-Holdout or Patient-Holdout. Through ensemble bootstrap aggregation and random subspace selection, individual decision trees draw upon complementary morphological features (e.g., S-wave in Slice 7 and early ventricular deflection in Slice 5 alongside the central R-peak). Consequently, its learned decision rules are robust to out-of-distribution subject morphologies, explaining why RF achieves a superior Macro F1-score (0.6342 vs. 0.5601) on unseen patients.

2. SVM Morphological Sensitivity ($r_s = 0.6682, p = 0.0246$):

While the SVM consistently identifies Slice 6 as the primary locus of importance, its overall slice ranking shifts more noticeably between validation protocols. The non-linear RBF kernel maps the 275-dimensional ECG signal into an infinite-dimensional feature space where support vector margins are highly sensitive to exact point coordinates. As a result, inter-patient baseline variations and electrical axis shifts induce greater variation in the SVM's relative feature attributions.

3. Cross-Model Consensus ($r_s = 0.7189$ and $r_s = 0.7414$):

Under both validation protocols, the SVM and Random Forest display moderate-to-strong rank agreement ($r_s > 0.71$). Those two fundamentally distinct learning algorithms. A margin-maximizing hyperplane kernel versus an ensemble of orthogonal axis-aligned decision trees converge on the same temporal prioritization confirms that QRS dominance is a genuine property of cardiac electrophysiology rather than an architectural artifact of a specific model.

Statistical Limitations of the Correlation Analysis

Several caveats must be acknowledged regarding the Spearman rank correlation analysis:

1. Small Sample Size ($N = 11$ points):

Each rank correlation was computed across only 11 data points (one per temporal slice). With such small degrees of freedom ($df = 9$), statistical power is restricted, and asymptotic p-values must be interpreted with caution. We therefore emphasize the magnitude of the rank correlation coefficients (r_s) as descriptive indicators of moderate-to-strong alignment rather than drawing strict binary conclusions from uncorrected significance thresholds.

2. Multiple Comparison Corrections:

Across the four statistical comparisons, applying a conservative family-wise Bonferroni correction ($\alpha_{adj} = 0.05/4 = 0.0125$) would cause the first comparison ($p = 0.0246$) and third comparison ($p = 0.0127$) to fall marginally outside the adjusted threshold.

3. Temporal Autocorrelation Across Adjacent Slices:

ECG time series inherently violate the assumption of statistical independence. Adjacent 25-sample slices share continuous physiological voltage trajectories and low-frequency baseline characteristics. This spatial and temporal autocorrelation effectively inflates the nominal degrees of freedom, further underscoring why reported p-values should be viewed as supplementary heuristics rather than definitive proof of independence.

Conclusion

This case study explored how effectively supervised machine learning algorithms, Support Vector Machines (SVM) and Random Forests (RF), can classify different types of cardiac arrhythmias using the MIT-BIH Arrhythmia Database. Alongside comparing model performance, the study examined how different validation strategies affect the reliability of results and used Time-Slice Permutation Feature Importance (PFI) to understand which parts of an ECG signal influence model predictions. The main focus was to address two important challenges in medical AI: ensuring that model performance reflects real-world generalisation and making model decisions easier to interpret.

1. Generalisation to Unseen Patients and the Risk of Data Leakage

Since heartbeats from the same patient can appear in both the training and testing sets, models may learn patient-specific ECG patterns rather than features that genuinely distinguish different arrhythmias. This can result in exceptionally high-performance during testing without guaranteeing that the model will work equally well for new patients.Patient-Holdout validation provides a more realistic evaluation by ensuring that the model is tested on patients it has never encountered during training. Although the overall accuracy remained relatively high (94.34% for SVM and 94.02% for RF), these results did not tell the whole story.A closer look at the evaluation metrics revealed an important limitation. Normal and Paced beats accounted for more than 82% of the test samples, meaning that the models could achieve high accuracy while still struggling to identify less frequent arrhythmias. This was reflected in the Macro Recall, which dropped from 0.95 to 0.55, and the Macro F1-score, which decreased from 0.82 to 0.56 for SVM.These findings highlight why accuracy alone is not sufficient when evaluating medical classification systems. Macro Recall and Macro F1-score provide a more balanced view of performance across different classes, particularly when dealing with imbalanced medical datasets. More importantly, evaluating models on completely unseen patients is essential for understanding whether they can generalise beyond the individuals represented in their training data.

2. Understanding the ECG Features Behind Model Predictions

The Time-Slice Permutation Feature Importance analysis provided further insight into how the models made their predictions. Rather than treating the ECG signal as a collection of unrelated numerical values, the analysis helped identify which regions of the heartbeat contributed most to classification.The most significant finding was the strong importance of Slice 6 (samples 125–149), corresponding to approximately -33 to +36 ms around the R-peak. This region contributed the largest share of predictive importance across the models and validation settings, with the SVM showing an

error increase of up to 55.5% when this region was disrupted. Slice 7, which covers the S-wave and early ST-junction, was also consistently important. This makes sense from a clinical perspective. The P-wave region (Slices 1–4), associated with atrial depolarisation, and the T-wave region (Slices 8–11), associated with ventricular repolarisation, showed smaller but non-zero contributions. Overall, the feature importance results suggest that the models relied primarily on clinically relevant regions of the ECG waveform rather than distributing their attention randomly across the signal.

3. Comparing Model Robustness and Clinical Potential

When comparing the two algorithms, Random Forest showed greater consistency across the different validation settings, achieving a Spearman rank correlation of 0.8856. It also achieved a higher Macro F1-score on unseen patients than SVM (0.6342 compared with 0.5601). One possible explanation is the way Random Forest combines predictions from multiple decision trees. Since individual trees can learn different patterns from the ECG waveform, the ensemble can make use of a wider range of temporal features. This may help it handle variations in ECG morphology between patients more effectively than the RBF-based SVM, which relies on its learned decision boundaries. Another interesting observation was the agreement between the two algorithms in their feature importance rankings. The cross-model Spearman correlations ranged from 0.72 to 0.74, indicating a reasonable level of agreement about which ECG regions were important

Overall Conclusion

In summary,
(1) accuracy is a misleading measure on this imbalanced data, and macro-averaged metrics on unseen patients are the honest way to judge the models; (2) both classifiers rely mainly on the QRS complex, which is consistent with how clinicians recognise most of these beat types, and the agreement between two very different models supports this finding; and (3) the random forest generalised somewhat better to unseen patients in our experiments, although this difference should be confirmed with repeated experiments. We therefore recommend reporting macro-averaged metrics under patient hold-out, together with an explanation of what the model uses.